Roll No. | | | | | | | | | | | | | **Total No. of Pages : 03**

Total No. of Questions : 13

B.Pharma. (Sem.-2)

BIOCHEMISTRY

Subject Code : BP-203T

M.Code : 74969

Date of Examination : 17-12-2022

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Multiple choice questions :

a) Amino acid residues in protein contain……..number of chiral centre.

A. One, B. Two, C. Three, D. Four

b) a-D-glucose is a …… of P-D-glucose

A. Anomer, B. Epimer, C. Enantiomer, D. A and C

c) Cholesterol is a ……

A. Derived lipid, B. Precursor of bile acid

C. Amphiphathic lipid D. All of the above

d) For, N-Ca bond, the bond angle resulting from rotation at Ca is labeled .

A. Φ, B. ψ C. δ, D. A and B

e) In the structure of DNA, pentose ring is in

A. B-Furanose form, B. 2' Endo conformation,

C. A and B both, D. Neither A nor B

f) Enzyme required for the oxidation of fatty acids are located in

A. Cytosol, B. Mitochondrial Matrix,

C. A and B both, D. Interstitial spaces

g) Formation of ATP in glycolysis is an example of

A. Substrate level phosphorylation, B. Oxidative phosphorylation,

C. Substrate level dephosphorylation D. Oxidative dephosphorylation

h) _____acid is essential fatty acid

A. Palmitic acid, B. Stearic acid,

C. Oleic acid, D. Linoleic acid

i) All transaminases require_____co-enzyme

A. PLP, B. FAD, C. NADH, D. CoASH

j) A group of ribonucleic acids that functions as bio-catalyst is known as—

A. Ribozymes, B. Isozymes, C. Co-enzymes, D. None of these.

SECTION-B

2. Describe various steps in glycogenesis. Comment on glycogen storage disease.

3. Describe the **De-Novo** synthesis of purine nucleotide. Comment on gout disease.

4. a) Discuss advantages of Line weaver Burke plot over Michalis-menten kinetics of enzyme action.

b) Explain allosteric regulation of enzymes.

SECTION-C

5. Describe common properties of α-amino acids.

6. Explain mechanism of ETC.

7. Give outline for conversion of cholesterol to bile acids.

8. Describe synthesis and biological significance of melatonin.

9. Describe various disorders of tyrosine metabolism.

10. Give outline for production of bilirubin by microsomal heme oxygenase system.

11. Write a short note on inhibitors of protein synthesis.

12. Give five important differences between DNA and RNA.

13. Give structure and biological importance of co-enzymes derived from vitamin B6.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.